

1. Introduction

In the weeks following the first project report, we focused primarily on the software aspect of the car. Building on the calibration work described previously, we concentrated on improving the car's perception system. Given our limited prior experience in training AI models, this phase also served as an opportunity to study, experiment with, and develop practical skills in AI training and model optimization. Despite several technical challenges, including image noise and dataset limitations, this process led to valuable insights and notable improvements in overall system performance.

2. Planned activities

Task/Week	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Lane detection experimentation						
Object detection experimentation						
Speed control						

3. Status of planned activities

3.1. Lane detection experimentation: Status: 70%.

For lane detection, we tested the algorithm developed in Progress 1 and implemented additional methods such as Canny edge detection and sliding windows (with HLS and sobel filtering). Initial results were unsatisfactory, as white noise from reflections on the running track disrupted lane detection and caused inaccurate trajectory estimation. Despite further camera calibration and parameter tuning, the issue persisted. We therefore developed a simple monitoring tool to validate the system and identified the transformer (bird-eye view perspective transformation) as the main source of the problem. Besides algorithm-based lane detections, we also tested AI-based lane detection. We trained the YOLO8 model for segmentation on lane datasets with 3 classes: left, middle, and right lanes. However, a new problem arised which was the inconsistent image resize from 2080 x 1088 to 640 x 480 compared to the original 640 x 480 taken directly from the camera. Therefore, we abandon the model and return to the raw algorithm.

3.2. Object detection experimentation: Status: 90%.

For object detection, we initially trained the YOLOv11 model on a custom, partially annotated dataset composed of samples from traffic-related datasets such as KITTI, MTSD, and BSTLD. The training process showed very slow convergence, which was later attributed to class imbalance. After rebalancing the dataset and retraining, we observed a slight improvement in detection accuracy. With the release of the YOLOv26 model and in recognition of the limitations of partially annotated data, we designed a new loss function to address this issue. Testing this new model resulted in a significant improvement in performance. We decided to train 5 prototypes with different parameters and then appended them to the car's camera, yielding acceptable result: while the best model faces accuracy issues on far objects, it performed well in proximity. Nonetheless, more efforts on traffic light detection should be made, as the model still only recognizes red light.

3.3. Speed control and steer control: Status: 70%.

For speed and steer control, we first use a smoothing technique called exponential moving average (70% old + 30% new). By implementing Pure Pursuit control theory, we set a lookahead distance to find the target destination along the virtual center road. To enhance tracking stability, the algorithm employs an adaptive lookahead mechanism where the search



radius is dynamically scaled linearly with the vehicle's current velocity, constrained within a defined range. The target trajectory is established by computing the polynomial intersection of this lookahead circle with the estimated lane center—derived either by averaging the coefficients of both lane boundaries or by applying a lateral offset when only a single boundary is detected. The steering angle is subsequently calculated using the geometric relationship taken from Pure Pursuit controller. Finally, the longitudinal control is coupled with the lateral requirements; the reference velocity is inversely modulated by the steering magnitude, forcing the vehicle to decelerate during high-curvature maneuvers to maintain traction and tracking fidelity.

4. Upcoming activities

1. Advanced Control: The team will refine the control architecture by implementing and tuning a Proportional-Integral-Derivative (PID) controller, conducting comparative analysis against the current Pure Pursuit method to minimize cross-track error. Concurrently, the longitudinal control framework will be advanced to incorporate semantic road classification, enabling the vehicle to execute adaptive velocity profiling based on specific road types (e.g., reducing speed for curved segments or complex textures). In the future, we may change to Model-Predictive-Control based on ROS2 framework with the hardware upgrade. The team will work on the Finite State Machine as well as several autonomous features of the car (Stopping at traffic signs, red lights and on encountering pedestrians).

2. Complex Manuever and Perception: The development focus will shift towards the expansion of the Finite State Machine (FSM) to govern autonomous intersection traversal logic. This includes augmenting the perception stack with a robust traffic sign recognition module, ensuring the vehicle can reliably identify and adhere to regulatory signals (Stop, Priority) and successfully negotiate signalized intersections.

3. Spring-clean: we will improve our working environment by upgrading the equipments and the overall atmosphere to prepare for the qualification video.